Day 72 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in General Science (4): ISRO's SpaDeX Docking, New TN Spaceport & Gaganyaan Progress

READING MATERIAL

Covers three connected, dated ISRO/space-technology developments: the SpaDeX docking demonstration, the new TN-based spaceport, and Gaganyaan's human-spaceflight progress.

Q: What did ISRO's SpaDeX mission achieve, and when?

A: On 16 January 2025, ISRO successfully docked two satellites (SDX-01 'Chaser' and SDX-02 'Target', 220 kg each) in space — India's first satellite-docking demonstration, making India the 4th country (after the USA, Russia, and China) to achieve this capability.

Q: Why does the SpaDeX docking matter for India's future space missions?

A: It is a technology demonstrator for future missions requiring docking, including Chandrayaan-4, the Bharatiya Antariksh Station (India's planned space station), and India's planned Human Moon Mission.

Q: Where is ISRO's new second spaceport being built, and what is it called?

A: The SSLV Launch Complex (SLC), at Kulasekarapattinam, a coastal village in Thoothukudi district, Tamil Nadu — spread over about 2,233 acres.

Q: Why is the Kulasekarapattinam location strategically chosen for launches?

A: It allows launches directly south over the Indian Ocean without crossing any landmass for thousands of miles — ideal for efficiently launching satellites into polar orbits.

Q: What is the SSLV Launch Complex's planned timeline and capacity?

A: Construction began 5 March 2025; the launch pad is planned for commissioning by August 2026, with the first launch by December 2026; the facility is estimated to support 20-25 launches per year.

Q: What is 'Gaganyaan-1', and what is it scheduled to carry?

A: Gaganyaan-1 (G1) is ISRO's first UNCREWED test flight under the Gaganyaan human-spaceflight programme, scheduled for March 2026, carrying 'Vyommitra' — a half-humanoid robot — to simulate astronaut conditions and gather life-support/environmental data.

Q: What is Gaganyaan's overall crewed-mission timeline?

A: After uncrewed test flights (G1, G2, G3) through 2026, India's first crewed spaceflight (H1) is planned for 2027, carrying 3 astronauts to low Earth orbit for up to 7 days.

MOCK TEST (WITH ANSWERS)

1. ISRO's SpaDeX mission successfully docked two satellites on:

A. 30 December 2024

B. 16 January 2025 ✓

C. 12 January 2025

D. 7 January 2025

Explanation: 16 January 2025. [medium, web-research]

2. With the SpaDeX docking, India became the ____ country to demonstrate satellite docking.

A. 2nd

B. 3rd

C. 4th ✓

D. 5th

Explanation: 4th, after the USA, Russia, and China. [medium, web-research]

3. ISRO's new second spaceport, the SSLV Launch Complex, is located at:

A. Sriharikota, Andhra Pradesh

B. Kulasekarapattinam, Tamil Nadu ✓

C. Thumba, Kerala

D. Bengaluru, Karnataka

Explanation: Kulasekarapattinam, Thoothukudi district, Tamil Nadu. [easy, web-research]

4. Launches from Kulasekarapattinam are strategically advantageous because they can launch directly south over the Indian Ocean, making them ideal for:

A. Geostationary orbit launches

B. Polar orbit launches ✓

C. Lunar transfer launches

D. Deep-space launches only

Explanation: Polar orbit launches. [medium, web-research]

5. 'Vyommitra', scheduled to fly on Gaganyaan-1 (March 2026), is:

A. A communication satellite

B. A half-humanoid robot ✓

C. A weather-monitoring probe

D. A cargo module

Explanation: A half-humanoid robot. [medium, web-research]

6. India's first crewed Gaganyaan spaceflight (H1) is planned for:

A. 2026

B. 2027 ✓

C. 2028

D. 2029

Explanation: 2027. [hard, web-research]

Part B – Aptitude & Mental Ability: Area

READING MATERIAL

Area is 1 of 10 named Aptitude sub-skills. Real exam evidence shows a strong preference for applied/real-world framing (fencing cost, carpet cost, crossroads, a grazing horse) over bare formula plug-ins — and frequently combines area with a real-world cost-per-unit calculation.

Q: Method — area of a triangle from coordinates: Find the area of the triangle with vertices (0,0), (2,0), (0,2).

A: When two sides lie along the axes (a right triangle at the origin), area = ½ × base × height = ½ × 2 × 2 = 2 sq. units. (The general coordinate formula $\frac{1}{2}|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)|$ also works, but recognizing the right-angle-at-origin shortcut is faster.)

Q: Real exam: A rectangular ground 80m × 60m has two equal-width cross roads (one parallel to each side). Their combined area is 675 sq.m. Find the road width.

A: If width=w, the two roads' combined area = $80w + 60w - w^2$ (subtracting the overlapping square where they cross, counted twice otherwise) = $140w - w^2$. Set equal to 675: $w^2 - 140w + 675 = 0$. Solving gives w=5 (the other root, 135, is too large to be physically valid for an 80×60 ground).

Q: Real exam: Find the perimeter of a semicircle of radius 28cm.

A: Perimeter of a semicircle = πr + 2r (the curved half plus the straight diameter edge). Using π≈22/7: (22/7×28) + (2×28) = 88 + 56 = 144cm. (Common mistake: forgetting to add the straight diameter edge, which would wrongly give just 88cm.)

Q: Real exam: A horse is tethered by a 14m rope to one corner of a 60m×42m rectangular field. How much area is left UNGRAZED?

A: The horse can graze a quarter-circle of radius 14m (since it's tied at a corner, with two field-edges as boundaries): grazed area = $\frac{1}{4}\times\pi\times14^2 = \frac{1}{4}\times(22/7)\times196$ = 154 sq.m. Total field area = 60×42 = 2520 sq.m. Ungrazed = 2520−154 = 2366 sq.m.

Q: Real exam: Area of a rhombus with diagonals 6cm and 8cm?

A: Area of rhombus = $\frac{1}{2} \times d_1 \times d_2$ = ½×6×8 = 24 cm^2.

Q: Real exam: If fencing a circular park at ₹5/metre costs ₹1,100 total, find the park's radius.

A: Total fencing length (= circumference) = ₹1,100 ÷ ₹5/m = 220m. Circumference = 2πr → 220 = 2×(22/7)×r → r = 220×7/44 = 35m.

Q: Real exam: Find the cost of carpeting a room 13m long, 9m wide, using a carpet 75cm wide, at Rs.8 per metre (of carpet length).

A: Room area = 13×9 = 117 m^2. Carpet width = 0.75m, so length of carpet needed = Room area ÷ carpet width = 117/0.75 = 156m. Cost = 156 × Rs.8 = Rs.1,248.

Q: Real exam (related, surface area of a 3D shape): The curved surface area of a right circular cylinder of height 7cm is 66 cm^2. Find its diameter.

A: CSA = 2πrh → 66 = 2×(22/7)×r×7 = 44r → r=1.5cm → diameter = 3cm.

MOCK TEST (WITH ANSWERS)

1. Find the area of the triangle with vertices (0,0), (5,0), (0,4).

A. 10 sq. units ✓

B. 20 sq. units

C. 9 sq. units

D. 18 sq. units

Explanation: Right triangle at origin: area = ½×5×4 = 10 sq. units. [easy, authored]

2. Find the perimeter of a semicircle of radius 21cm (use π≈22/7).

A. 66 cm

B. 108 cm ✓

C. 132 cm

D. 96 cm

Explanation: Perimeter = πr+2r = (22/7×21)+(2×21) = 66+42 = 108cm. [medium, authored]

3. Find the area of a rhombus with diagonals 10cm and 12cm.

A. 60 cm^2 ✓

B. 120 cm^2

C. 30 cm^2

D. 100 cm^2

Explanation: Area = ½×10×12 = 60 cm^2. [easy, authored]

4. Fencing a circular garden at ₹4 per metre costs ₹880 in total. Find the garden's radius (use $\pi \approx 22/7$).

A. 28 m

B. 35 m ✓

C. 22 m

D. 14 m

Explanation: Circumference = 880/4 = 220m. r = 220×7/(2×22) = 1540/44 = 35m. [medium, authored]

5. Find the cost of carpeting a room 10m long and 8m wide with a carpet 80cm wide, at Rs.10 per metre.

A. Rs.800

B. Rs.1,000 ✓

C. Rs.1,200

D. Rs.960

Explanation: Room area = 80 m^2. Carpet length needed = 80/0.8 = 100m. Cost = 100×Rs.10 = Rs.1,000. [medium, authored]

6. A horse is tethered by a 7m rope to a corner of a 20m×15m rectangular field. Find the grazed area (use $\pi \approx 22/7$).

A. 38.5 m^2 ✓

B. 44 m^2

C. 49 m^2

D. 77 m^2

Explanation: Quarter circle of radius 7m: ¼×(22/7)×49 = 38.5 m^2. [medium, authored]

Part C – General English: Reading Comprehension: Governance & Civic Passages (Set 2)

READING MATERIAL

Longer RC passages (5+ lines) often carry 2-3 questions each, testing different skills from a single passage: the main function/purpose, an inference about implications, and vocabulary-in-context (what a specific word means as used in that passage, which may differ from its most common dictionary meaning).

Q: Passage: "'E-sevai centers', a cornerstone of Tamil Nadu's e-governance, streamlines citizen access to essential services. These centers facilitate online delivery of documents like birth certificates and land records, reducing bureaucratic delays... The long-term success of e-sevai depends on bridging the digital divide and maintaining a robust infrastructure." What is the primary function of e-sevai centers, per the passage?

A: To facilitate online delivery of essential government services (e.g., birth certificates, land records) — not to provide free internet access, train employees, or manage databases, which are side details, not the primary function.

Q: From the same E-Sevai passage: what does the passage say the long-term sustainability of e-sevai primarily relies on?

A: Bridging the digital divide and maintaining a robust infrastructure (stated explicitly in the passage's final sentence) — not simply increasing the number of physical centers or focusing only on urban areas.

Q: From the same E-Sevai passage: in context, what does 'streamline' mean?

A: Simplify — vocabulary-in-context questions ask for the meaning AS USED in that specific passage; here 'streamlines citizen access' clearly means 'simplifies', not complicate/delay/increase cost.

Q: Passage: "Reforestation projects restore degraded ecosystems. They increase biodiversity and improve air quality. They also enhance water retention. These efforts support long-term environmental sustainability." Which statements best support the claim that reforestation is beneficial?

A: (b) Reforestation projects restore degraded ecosystems; (c) Reforestation projects improve air quality; (d) Reforestation projects enhance water retention. [NOT (a) — 'decrease biodiversity' directly contradicts the passage, which says reforestation INCREASES biodiversity.]

Q: Scenario: "You are a newly appointed Civil Servant in the Department of Climate Change, asked to communicate the department's initiatives to the public." Which collocations would be most effective? (a) to spread climate misinformation (b) to raise public awareness about climate change impacts (c) to minimize the urgency of climate action (d) to engage in stakeholder consultations

A: (b) and (d) — 'raise public awareness' and 'engage in stakeholder consultations' are appropriate, professional communication collocations for a civil servant; (a) and (c) work against the department's own goals, so they're never the right communication strategy regardless of phrasing quality.

MOCK TEST (WITH ANSWERS)

1. Passage: "'E-sevai centers', a cornerstone of Tamil Nadu's e-governance, streamlines citizen access to essential services. These centers facilitate online delivery of documents like birth certificates and land records..." What is the primary function of e-sevai centers, per the passage?

A. To provide free internet access to rural population

B. To facilitate online delivery of essential government services ✓

C. To train government employees in digital literacy

D. To manage state-level databases

Explanation: To facilitate online delivery of essential government services. [easy, question-bank-2025]

2. Per the same E-Sevai passage, the long-term sustainability of e-sevai primarily relies on:

A. Increasing the number of physical e-sevai centers

B. Focussing solely on urban areas

C. Bridging the digital divide and maintaining a robust infrastructure ✓

D. Reducing the number of available online services

Explanation: Bridging the digital divide and maintaining a robust infrastructure. [medium, question-bank-2025]

3. In the context of the E-Sevai passage, 'streamline' means:

A. Complicate

B. Simplify ✓

C. Delay

D. Increase cost

Explanation: Simplify. [easy, question-bank-2025]

4. Passage: "Reforestation projects restore degraded ecosystems. They increase biodiversity and improve air quality. They also enhance water retention." Which statements best support the claim that reforestation is beneficial? (a) decrease biodiversity (b) restore degraded ecosystems (c) improve air quality (d) enhance water retention

A. (b), (a), (c)

B. (b), (d), (a)

C. (b), (c), (d) ✓

D. (a), (b), (c)

Explanation: (b), (c), (d) — (a) contradicts the passage. [medium, question-bank-2025]

5. Scenario: a Civil Servant in the Dept. of Climate Change must communicate initiatives to the public. Which collocations are most effective? (a) spread climate misinformation (b) raise public awareness about climate change impacts (c) minimize the urgency of climate action (d) engage in stakeholder consultations

A. (a) and (c)

B. (b) and (d) ✓

C. (b) and (c)

D. (a) and (d)

Explanation: (b) and (d). [medium, question-bank-2025]